

Week 10 Workshop Solutions

1. AR(2)-GARCH(1,1) model

$$\begin{aligned}r_t &= c + \phi_1 r_{t-1} + \phi_2 r_{t-2} + \varepsilon_t \\ \varepsilon_t &= \sigma_t u_t, \quad u_t \sim i.i.d.N(0, 1) \\ \sigma_t^2 &= \alpha_0 + \alpha_1 \varepsilon_{t-1}^2 + \beta_1 \sigma_{t-1}^2\end{aligned}$$

with $\alpha_0 > 0, \alpha_1 \geq 0$ and $\beta_1 \geq 0$. Also $\alpha_1 + \beta_1 < 1$.

2. (a)

```
> fit_garch_m@fit$matcoef
      Estimate Std. Error   t value    Pr(>|t|)
mu      0.04687983 0.014651678   3.1996223 1.376078e-03
archm    0.01539739 0.021515733   0.7156338 4.742175e-01
omega    0.00789248 0.001855065   4.2545561 2.094642e-05
alpha1   0.09242559 0.009478617   9.7509572 0.000000e+00
beta1    0.90025195 0.009893088  90.9980691 0.000000e+00
```

$$\begin{aligned}r_t &= 0.047 + 0.015\hat{\sigma}_t^2 + \hat{\varepsilon}_t \\ \hat{\sigma}_t^2 &= 0.008 + 0.0924\hat{\varepsilon}_{t-1}^2 + 0.900\hat{\sigma}_{t-1}^2\end{aligned}$$

(b). The null and alternative hypotheses are

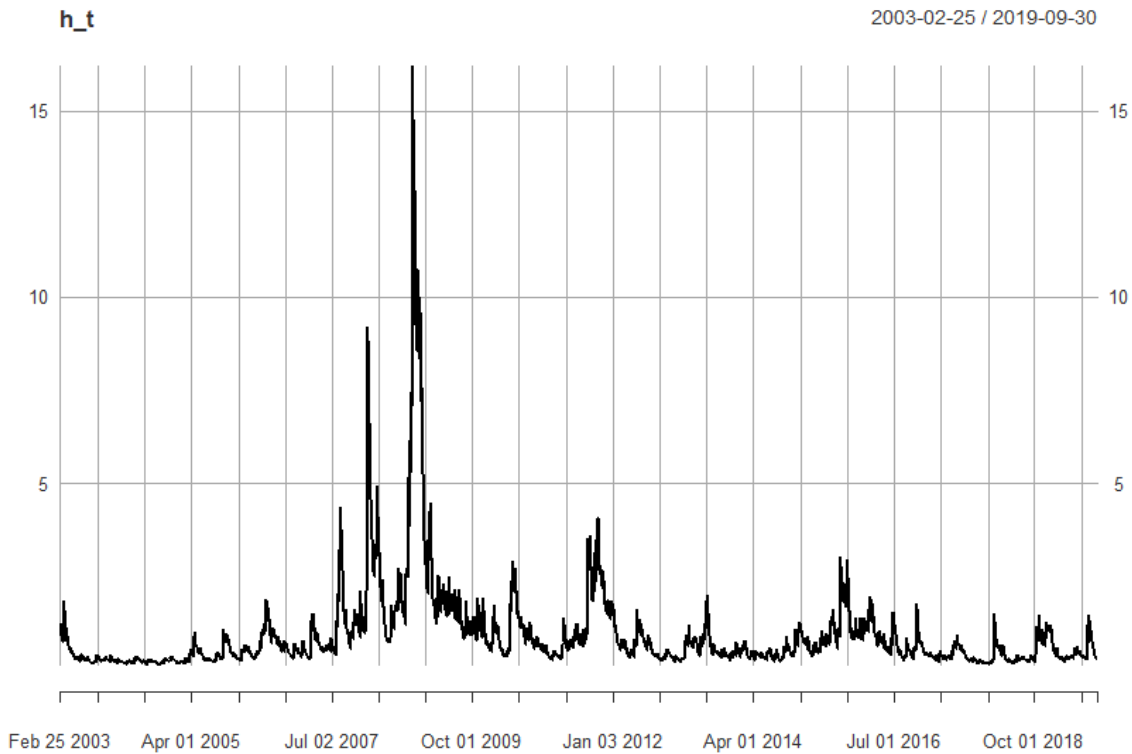
$$\begin{aligned}H_0 &: c = 0 \\ H_A &: c \neq 0\end{aligned}$$

The test statistic is $t = 3.1996223$, which has a p-value of 0.0014; therefore reject the null hypothesis at the 5% level and conclude the model should contain an intercept in the mean equation.

Evidence of a non-zero intercept is inconsistent with the Merton ICAPM as this model predicts a proportionate relationship between expected returns and the variance, where the proportionality factor is the relative risk aversion parameter γ .

(c). The estimate of the relative risk aversion parameter γ is $\hat{\gamma} = 0.015$. Since the estimate is positive, the point estimate suggests a positive relationship between expected returns and risk. However, the estimate is statistically insignificant, with p-value = 0.474 > 0.05. Therefore, there is insufficient statistical evidence of a positive risk-return trade-off.

(d). A plot of the estimated conditional variance $\hat{\sigma}_t^2$ is given below.



The figure shows the big increase in uncertainty around the time of the Global Financial Crisis (GFC) in 2008-09. In fact there are two bursts of volatility: one at the start of 2008 and another nearer to the end of 2008.

3. A standard likelihood ratio test is appropriate for nested models. Since GJR-GARCH and EGARCH are non-nested competing models, we should not use a standard LR test to compare them.

```
> # information criteria AIC
> Ics.gjr11 <- infocriteria(fit_gjr11)

> Ics.gjr11

Akaike      2.353870
Bayes       2.361404
Shibata     2.353867
Hannan-Quinn 2.356533

> Ics.egarch11 <- infocriteria(fit_egarch11)

> Ics.egarch11

Akaike      2.346901
Bayes       2.354435
Shibata     2.346898
Hannan-Quinn 2.349564
```

The EGARCH(1,1) model is preferred because it has the lower AIC and lower Bayesian information criterion, reported as “Bayes” in the R output.

4. The fitted ARCH(1) and ARCH(4) models

```
> fit_arch1@fit$matcoef
      Estimate Std. Error  t value    Pr(>|t|)
mu      0.05356406  0.01254761  4.268867 1.964685e-05
omega   0.58125043  0.01821456 31.911311 0.000000e+00
alpha1  0.42250244  0.03337468 12.659372 0.000000e+00
> fit_arch1@fit$LLH
[1] -5550.129

> fit_arch4@fit$matcoef
      Estimate Std. Error  t value    Pr(>|t|)
mu      0.0650848  0.01103490  5.898084 3.677457e-09
omega   0.2727968  0.01445919 18.866674 0.000000e+00
alpha1  0.1730885  0.02183225  7.928109 2.220446e-15
alpha2  0.1993542  0.02170472  9.184835 0.000000e+00
alpha3  0.1548500  0.01923212  8.051637 8.881784e-16
alpha4  0.1888029  0.02205846  8.559205 0.000000e+00
>
> # log-likelihood
> fit_arch4@fit$LLH
[1] -5173.618
```

The Likelihood ratio test works when one model is nested in another. We have an ARCH(1) and an ARCH(4) model:

$$\begin{aligned}\sigma_t^2 &= a_0 + a_1 \varepsilon_{t-1}^2, \\ \sigma_t^2 &= a_0 + a_1 \varepsilon_{t-1}^2 + \dots + a_4 \varepsilon_{t-4}^2.\end{aligned}$$

The ARCH(4) is the *unrestricted* model.

The ARCH(1) is just a *restricted* ARCH(4) with $a_2 = a_3 = a_4 = 0$.

Tested hypothesis:

$$\begin{aligned}H_0 &: a_2 = a_3 = a_4 = 0, \\ H_1 &: \text{At least one } a_i \neq 0 \text{ for } i = 2, \dots, 4.\end{aligned}$$

Test Statistic:

$$\begin{aligned}LR &= 2 \left[L_T \left(\hat{\theta}_{UR} \right) - L_T \left(\tilde{\theta}_R \right) \right] \\ &= 2 \left(-5173.618 - (-5550.129) \right) \\ &= 753.02\end{aligned}$$

The log-likelihood of the unrestricted model is at least as large as that of the restricted model and so LR is positive.

Distribution under the Null: LR has a χ_3^2 distribution if H_0 is true.

Degrees of freedom = number of restrictions = 3

The critical value is $\chi_3^2(0.05) = 7.815$ and so we reject the null hypothesis. Thus, the ARCH(4) fits the data better.